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PAPER

Computer-aided diagnosis of early-stage Retinopathy of Prematurity in neonatal fundus images using artificial intelligence

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Abstract

Retinopathy of Prematurity (ROP) is a retinal disorder affecting preterm babies, which can lead to permanent blindness without treatment. Early-stage ROP diagnosis is vital in providing optimal therapy for the neonates. The proposed study predicts early-stage ROP from neonatal fundus images using Machine Learning (ML) classifiers and Convolutional Neural Networks (CNN) based pre-trained networks. The characteristic demarcation lines and ridges in early stage ROP are segmented utilising a novel Swin U-Net. 2000 Scale Invariant Feature Transform (SIFT) descriptors were extracted from the segmented ridges and are dimensionally reduced to 50 features using Principal Component Analysis (PCA). Seven ROP-specific features, including six Gray Level Co-occurrence Matrix (GLCM) and ridge length features, are extracted from the segmented image and are fused with the PCA reduced 50 SIFT features. Finally, three ML classifiers, such as Support Vector Machine (SVM), Random Forest (RF), and *k*- Nearest Neighbor (*k*-NN), are used to classify the 50 features to predict the early-stage ROP from Normal images. On the other hand, the raw retinal images are classified directly into normal and early-stage ROP using six pre-trained classifiers, namely ResNet50, ShuffleNet V2, EfficientNet, MobileNet, VGG16, and DarkNet19. It is seen that the ResNet50 network outperformed all other networks in predicting early-stage ROP with 89.5% accuracy, 87.5% sensitivity, 91.5% specificity, 91.1% precision, 88% NPV and an Area Under the Curve (AUC) of 0.92. Swin U-Net Convolutional Neural Networks (CNN) segmented the ridges and demarcation lines with an accuracy of 89.7% with 80.5% precision, 92.6% recall, 75.76% IoU, and 0.86 as the Dice coefficient. The SVM classifier using the 57 features from the segmented images achieved a classification accuracy of 88.75%, sensitivity of 90%, specificity of 87.5%, and an AUC of 0.91. The system can be utilised as a point-of-care diagnostic tool for ROP diagnosis of neonates in remote areas.

1. Introduction

Retinopathy of Prematurity (ROP) is a vision disorder in premature neonates, resulting in irreversible vision loss if left untreated [1]. Neonates born at a gestational age (GA) less than 32 weeks, with a birth weight (BW) less than 1500 grams are primarily affected, while the other risk factors include supplemental oxygen therapy, pulmonary complications, anaemia and intraventricular haemorrhage. Recent technical innovations in medicine have improved the survival rates of preterm neonates, which in turn increase the number of neonates affected by ROP [2]. Nearly 50000 neonates are affected with ROP-related irreversible visual impairment globally every year, with the maximum population from Latin America, Southeast Asia, and Eastern Europe [3]. In 2010, 10% of the worldwide estimate of infants affected by blindness due to ROP are from India [4].

According to 3rd International Classification of Retinopathy of Prematurity (ICROP) [5], ROP can be characterised mainly into (a) Zone - Vascularization Location; (b) Plus and Preplus- dilation and tortuosity of the posterior retinal vasculature; (c) Acute disease—Includes Stage 1 (Demarcation lines), Stage 2(Ridges) and Stage 3 (Extra-retinal Neovascular Proliferation);

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(d) Aggressive ROP - tempo of disease and appearance of vascular abnormalities; (e) Retinal Detachment-Includes Stage 4 (Partial Retinal Detachment) and Stage 5 (Total Retinal Detachment). Early-stage ROP indicates the presence of stage 1, stage 2 or stage 3 ROP. A thin demarcation line at the vascular and avascular retinal junction characterises stage 1 ROP. This demarcation line evolved into a ridge at stage 2 with characteristic height and width. In stage 3, extra-retinal neovascular proliferation arises from this ridge and extends to the vitreous.

ROP is typically screened by binocular indirect ophthalmoscopic assessment, which is challenging to perform in neonates and requires expertise. Jalali *et al* emphasise that delaying the first screening of ROP results in the progression of the disease and reduces the success of the treatments [6]. The diagnosis of the early-stage ROP is vital in providing optimal therapy for the neonates, which can prevent the onset of severe-stage ROP. Computer-aided automated diagnosis of this early-stage ROP is significant in relatively screening a large population of infants in rural areas with telemedicine assistance.

The proposed study is focused on classifying the early-stage ROP from normal neonatal retinal images using a standalone CNN. The study aims to (i) Segment the ridges and demarcation lines from ROP images using transformer-based Swin-UNet; (ii) Extract the Scale Invariant Feature Transform (SIFT) descriptors from the segmented ridges and demarcation lines and classify them using ML classifiers such as Support Vector Machine (SVM), Random Forest (RF), and *k*-Nearest Neighbour (*k*-NN); (iii) The raw neonatal fundus images are classified into early-stage ROP and normal using six pre-trained networks such as DarkNet19, VGG16, MobileNet, Efficient Net, ShuffleNet and ResNet50. The novelty of the proposed work is three-fold. Firstly, the study segments thin demarcation lines and ridges from early-stage ROP images using a state-of-the-art Swin Transformer-based U-shaped Encoder and Decoder architecture with skip connections for local and global semantic feature learning. Secondly, in the Swin block, encoder with shifted windows extracts context features from the neonatal retinal image patches, which has not been done in previous works. Thirdly, the study examines the effectiveness of the standalone pre-trained classifiers in predicting early-stage ROP from neonatal fundus images.

Significant contributions of proposed work are given below:

- The study achieves precise segmentation of fragile demarcation lines and ridges from neonatal fundus images using the state-of-the-art transformer Swin-UNet, which effectively utilises global and long-range semantic information.
- The high-level SIFT features with 2000 descriptors, six GLCM features and ridge length are extracted from the demarcation lines and ridges segmented.
- The dimensionality of extracted features is reduced to 50 using Principal Component Analysis (PCA). They are fused with GLCM and ridge length features to obtain 57 features and are classified using the ML classifiers such as SVM, RF and *k*-NN.
- The ML classification involving segmentation, feature extraction, and dimensionality reduction is compared with classification using six standalone pre-trained networks: DarkNet19, VGG16, MobileNet, EfficientNet, ShuffleNet V2, and ResNet50.

2. Related works

The bloom of Artificial Intelligence (AI) has drastically improved several medical image diagnosis [7, 8]. The utilisation of machine learning (ML) classifiers and Convolutional Neural Networks (CNN) in the diagnosis of breast cancer [9], brain tumour [10], hematoma [11], retinal diseases such as diabetic retinopathy [12], glaucoma [13] etc, have been carried out in recent years.

Rao *et al* performed binary classification of ROP from normal fundus images [14]. The CNN utilised is the pre-trained EfficientNet-B0 model to classify 227,326 images of stages 1–3 from the KIDROP tele-ROP screening program. The classification accuracy obtained is 91.29%, with 91.46% sensitivity and 91.22% specificity. The authors utilised a single pre-trained model that could be compared with the performance of other lighter CNNs. Li *et al* designed an automated system for multistage classification of stages 1–3 using three datasets containing 18,827 images [15]. The authors segmented the demarcation lines and ridges using U-Net and classified them using the pre-trained DenseNet model. Although the system achieves an accuracy of 95.33%, its complexity is high as three high-performing CNNs are utilized to handle the image dataset.

Peng *et al* performed multistage ROP classification in 635 retinal fundus images [16]. The high-level features are extracted from the raw images using three-stream networks: ResNet18, DenseNet121, and EfficientNetB2. These features are fused and classified using an ordinal classification strategy with an accuracy of 98.27%, sensitivity of 90.55%, and precision of 90.92%. However, the system with three pre-trained parallel networks is computationally expensive. Ding *et al* designed an automated system to classify stages 1–3 ROP using an ensembling approach [17]. The authors identify the demarcation lines and ridges using the Mask RCNN network and classify them using Resnet-101 CNN. The classification accuracy obtained by the ensemble network is 67%, which could be improved.

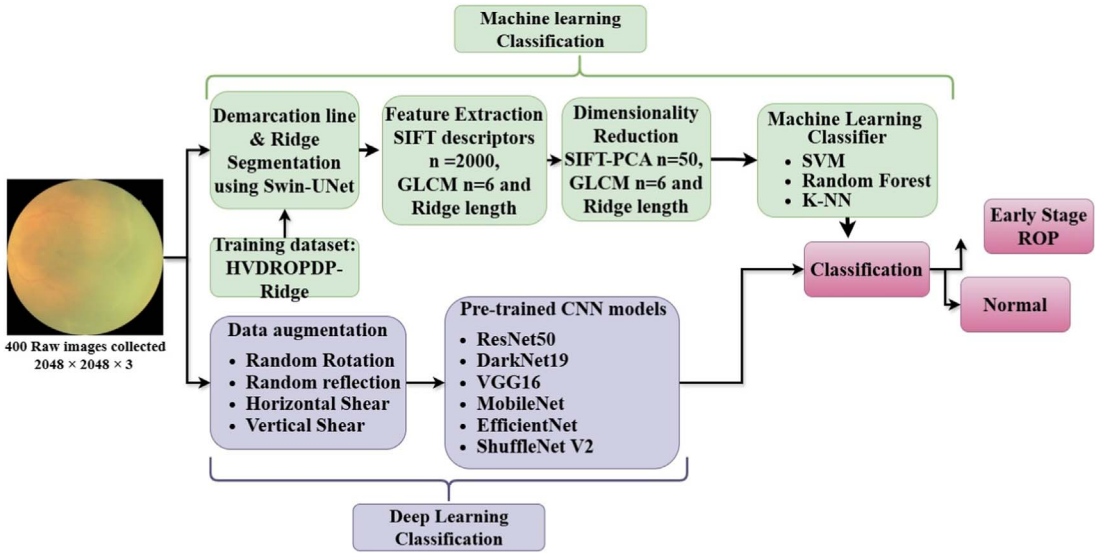


Figure 1. The schematic diagram of the proposed network for ROP classification.

The automated classification system designed by Tong *et al* utilised Faster RCNN to classify the stage of ROP using 38,895 fundus images [18]. Even though the system yielded a maximum classification accuracy of 90.3%, it is complex with more parameters. Mulay *et al* segmented the demarcation line and ridges of stage 2 using the Mask R-CNN network [19]. A total of 220 images were normalised, and regions of interest were generated using ResNet-101-FPN and RPN. The visual features are extracted from these generated regions and are combined into a single bounding box to provide the segmented result. The model attained 88% segmentation accuracy for early-stage ROP. Previous research works [15–19] involve more than one CNN to predict the early-stage ROP, which is computationally expensive.

3. Methodology

3.1. Data collection

The publicly available dataset HVDROPDB-RIDGE [20] is used to train the Swin U-Net in segmentation. The dataset consists of 50 neo-images and their corresponding masks of demarcation lines and ridges of early-stage ROP infants of 26–36 weeks GA and 760–3000 g BW, graded by ROP experts. The images used for classification were obtained retrospectively from SRM Medical College and RC, Kattankulathur, in association with Aravind Hospital, Poonamalle from July 2020 to December 2023. The fundus images of the neonates without eye disorders, who are registered in ROP diagnosis, are obtained for the proposed work. The retina of the neonates was captured by 3 Nethra Neo (Forus Health) digital wide field paediatric fundus camera with a 120° FOV and 2048 × 2048 resolution. The study includes 400 images (200 Early-stage ROP and 200 Normal) graded by qualified paediatric ophthalmologists. The images were captured from neonates of mean GA 31.5 ± 1.3 weeks with mean BW as 1.56 ± 0.32 kg. The research was conducted in accordance with the principles embodied in the Declaration of Helsinki and in accordance with local statutory requirements. All participants gave written informed consent to participate in the study. The proposed work is permitted by Institutional Ethics Committee, SRM Hospital & Medical College and Research Centre, with clearance number 8241/IEC/2022.

3.2. Proposed workflow

The overall flow of the current study in early-stage ROP classification is illustrated in figure 1. The transformer-based Swin-UNet is trained and validated to segment the demarcation lines and ridges from neonatal fundus images using the HVDROPDB-RIDGE dataset. 400 images collected are then provided to the Swin-UNet CNN to segment the early-stage demarcation lines and ridges, from which 2000 SIFT descriptors were extracted. The dimensions of the obtained features were reduced to 50 by PCA to obtain the maximum correlated components with reduced complexity. Fifty principal components 6 GLCM and ridge length are classified into normal and early-stage ROP using the ML classifier like SVM and compared with the performance of the other two classifiers such as RF and *k*-NN. The entire process of segmentation, feature extraction, dimensionality reduction and classification using ML is bypassed by deep learning classification. The raw images are augmented to 2000 (1000 normal and 1000 early-stage ROP) using four geometrical augmentation techniques such as random rotation, reflection, horizontal and vertical shear. The augmented images were classified into normal and early-stage ROP using six pre-trained CNNs, namely

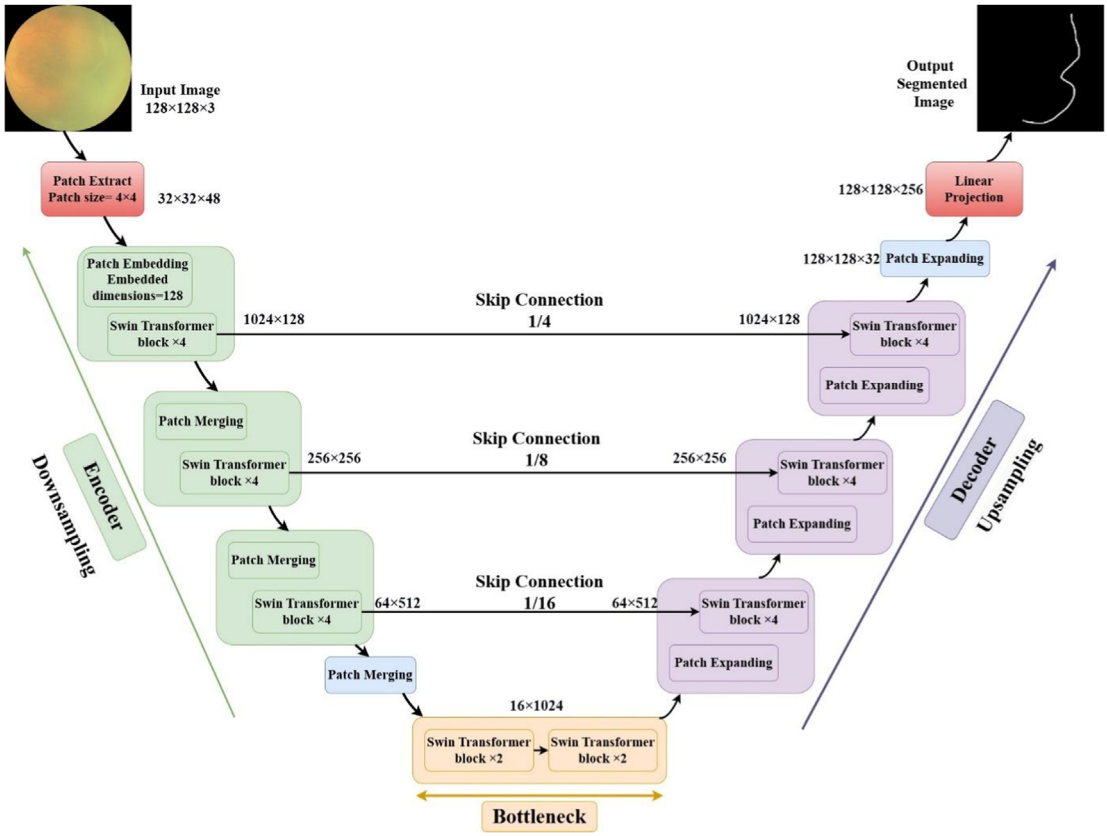


Figure 2. Architecture of swin U-Net network.

DarkNet19, VGG16, MobileNet, EfficientNet, ShuffleNet V2, and ResNet50.

3.3. Ridge segmentation

Recently, many researchers have utilized U-shaped CNNs with skip connections in medical image segmentation. These U-Nets have an encoder-decoder architecture that extracts deep features with larger receptive fields to yield precise segmentation [21]. However, the convolution-based U-Net could not extract long-range semantic information from the medical images [22]. The effectiveness of transformers in natural language processing motivates its use in image recognition and classification [23]. The Vision transformer [ViT] converts the input images into patches using linear embeddings and recognises the image features using added positional embeddings and a self-attention mechanism [24]. The hierarchical model Swin transformer uses shifted windows [25], which improves the efficiency of computation by non overlapping local and cross window connections. U-Net architecture with encoder and decoder with Swin transformer as the backbone is the state-of-the-art segmentation network in medical image analysis. The architecture of Swin U-Net utilised in segmentation of ridges is given in figure 2.

The Swin U-Net architecture has three main structures, namely, downsampling encoder, bottleneck and upsampling decoder. The retinal fundus images are resized to dimensions 128 × 128 × 3 and are converted into 1024 × 3 patches of size 4 × 4. These patches are embedded into tokens of size 1024 × 128 and are provided to the encoder of the architecture. The encoder has three Swin transformer blocks, with each having four Swin Transformers to provide hierarchical feature representations. The Swin transformer extracts the long-range and contextual features from an image using window based multihead self attention and the shifted window based multihead self attention modules followed by residual connection and a two-layer multi-layer perceptron module with GELU non-linearity. In the Swin transformer block, the first transformer is built with four attention heads, while the other three have eight attention heads. Each Swin transformer block is followed by a patch merging block, which downsamples the embedded patches by reducing the number of patches to half and doubling the embedded dimensions.

The bottleneck of Swin U-Net architecture consists of two Swin Transformer blocks, which learn the extracted deep feature representation, keeping its dimensions unaltered. The extracted contextual deep features are provided to the decoder, which consists of three Swin transformer blocks with four Swin transformers each. Each of these blocks is followed by a patch-expanding layer that upsamples the feature map and concatenates with the features obtained from the encoder using skip connections to decrease the spatial

information loss due to downsampling. Finally, the upsampled feature maps are fed to the linear projection layer, which generates the output pixel-level segmented image from the extracted features. The dataset HVDROPDB-RIDGE containing 50 neo images and their corresponding ridge/demarcation line annotations are used to train and test the Swin UNet architecture. 80% of the images (40) is employed for training, 10% of the images (5) is employed for validating, and the remaining 10% (5 images) is used for testing the network. The network is trained for 300 epochs using Categorical Cross entropy as the loss function, and adam as the optimiser with a learning rate of $1 \times$ e-4.

3.4. Feature extraction and dimensionality reduction

400 neonatal fundus images collected from the hospital are segmented using the trained Swin UNet. The SIFT algorithm is used to extract robust scale-invariant and rotation-invariant features from the 400 segmented images. The segmented images are converted into scale space images of varying scales, and the keypoints are estimated based on the Difference of Gaussian (DoG) [26]. The optimal keypoints are then chosen according to the local maxima and minima of the segmented ridges. The low-contrast key points that are positioned near the edges are eliminated. Finally, 2000 distinctive key point descriptors that provide information about the localisation of the ridges or demarcation lines of neonatal fundus images are extracted.

2000 SIFT features extracted from a single image tremendously increase the complexity of the computation when processed in neural networks. Hence, to reduce the 'curse of dimensionality', PCA is used. PCA effectively reduces the input features into a smaller set of features without losing the original information from the features [27]. PCA estimates the maximum variance in the given features and estimates the set of orthogonal axes called principal components. These principal components have maximum variance, resulting in having maximum information from the input features but in reduced dimensions. 2000 SIFT features extracted from the segmented ridges or demarcation lines are reduced to 50 principal components using PCA, thereby significantly reducing the computational complexity in the subsequent steps.

In order to extract the ROP-specific image features, texture-based GLCM features and ridge length are utilised. GLCM features are obtained based on a co-occurrence matrix that captures the frequency with which pixel pairs with specific intensity values appear together at a defined spatial relationship in the image. In the current study, six GLCM features such as contrast, correlation, energy, homogeneity, entropy and dissimilarity at 45°-pixel intervals and the length of the segmented ridge are extracted. These seven features, along with the PCA reduced 50 SIFT features, are fused to get 57 length feature vector in order to classify the neonatal retinal images.

3.5. Machine learning classification

ML classifiers are used to solve real-world classification and regression problems using organised data. From the total 400 images, 80% of the data (320 images) is employed for training, and the 20% (80 images) for testing. The dimensionally reduced 50 principal components and 7 features from GLCM and ridge length from each image are provided as input features for the ML classifier, and the resulting target is the binary output whether the image is of early ROP or normal.

3.5.1. SVM classifier

SVM is a supervised learning algorithm aimed at finding the optimal hyperplane that can classify the features of the target classes in the feature space [28]. The hyperplane is chosen such that it separates the target classes with the largest margin. The hyperparameters tuned to classify the retinal images are the C as 1.0 with squared l2 penalty, Radial Basis Function (RBF) as the kernel, and the gamma value is set to vary based on the variance between the input features.

3.5.2. Random forest classifier

The RF classifier, a supervised learning algorithm, creates an ensemble of decision trees to result in a single accurate outcome [29]. The decision trees are built using a random subset of the data to estimate the randomness of the features. The resulting randomness causes variability between each decision tree, which in turn improves the accuracy of the prediction without overfitting. Each tree gives one target class as the prediction. The target class that is chosen by maximum trees is the final classification outcome, showcasing the robustness of the RF Classifier. The total number of trees in each forest is set to 10, with the minimum samples needed to separate a given internal node is chosen to be 2, and Gini index as the impurity criteria to measure the quality of a split, further reinforcing its reliability in retinal image classification.

3.5.3. k-nearest neighbour classifier

k-NN, a supervised algorithm, operates without any underlying assumptions on data distribution [30]. It aims to find the k number of nearest instances from the particular instance based on the chosen distance metrics. The target class of the particular instance is given as the estimated result of the maximum number of members in a particular group. For this particular classification problem, the number of neighbours to estimate is tuned to 10, showcasing the *k*-NN's adaptability. For this particular classification problem, the number of neighbours to estimate is tuned to 10 with the Euclidean distance as the distance metric.

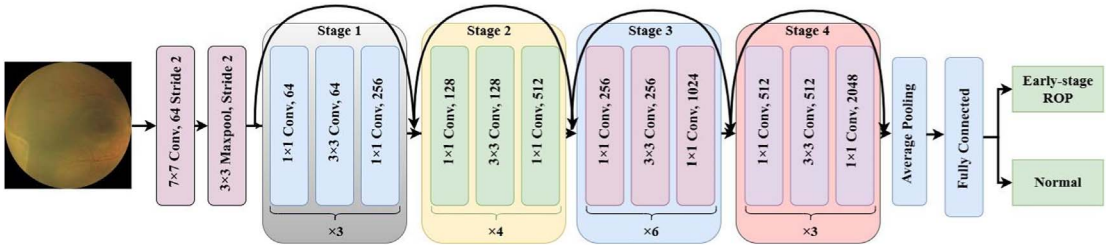


Figure 3. The architecture of ResNet50 CNN.

3.6. Pre-trained CNNs

Pre-trained CNNs are deep learning networks that are trained previously using the 1000 class images stored in the ImageNet [31] Database. These networks can be used to solve new classification problems using their existing knowledge and pre-trained weights by a process called transfer learning. It is done by freezing the initial layers of the CNNs without altering their weights and replacing the final classification layer based on the problem to be solved. Since detecting early-stage ROP is a binary classification problem, the final layer is altered by one neuron and replacing softmax with a sigmoid activation function. 400 collected raw images are augmented to 2000 images (1000 normal and 1000 early-stage ROP) using four geometrical augmentation techniques such as random rotation, reflection, horizontal and vertical shear. The augmented images are classified into normal and early-stage ROP using six pre-trained networks, namely DarkNet19, VGG16, MobileNet, EfficientNet, ShuffleNet V2, and ResNet50, using a 2-fold cross-validation method.

He *et al* [32] developed a fifty-layered deep pre-trained CNN ResNet50 with characteristic skip connections. It has 48 convolutional layers of varying filters, 1 max-pooling and 1 average-pooling layer. The gated recurrent skip connections between these layers reduce the computational time of ResNet50 compared to other networks, such as VGG16. The architecture of the ResNet50 Network is illustrated in figure 3. The raw neonatal fundus image is provided as input to the initial convolutional layer with 64 filters of size 7×7 at stride 2, which is then followed by a maxpooling layer of stride 2. The resulting feature map is given to four stages of convolution. Each stage has three convolutional layers with 1×1, 3×3, and 1×1 sized filters respectively.

The first and last stage of convolutions are repeated thrice, the second stage is repeated 4 times and the final stage is repeated 6 times. If a particular stage does not have the features corresponding to the target, the skip connections are activated and the particular stage is skipped. Finally, the feature map is given to the global average pooling layer and the classification layer with one neuron of sigmoid activation function, where the images are classified into normal or early-stage ROP.

Table 1. Hyperparameters tuned for ROP prediction.

Classifier	Hyperparameter	Tuned value
SVM	C	1.0
	Kernel	RBF
	Penalization	Squared l2 penalty
Random Forest	Number of trees	10
	Minimum samples needed	2
	Impurity criteria	Gini index
k-NN	Number of neighbours k	10
	Distance metric	Euclidean distance
PCA	Number of components	5
Pre-trained CNNs	Optimizer	SGDM
	Batch size	8
	Learning rate	$1 \times$ e-8
	Epochs	10

All six pre-trained networks are trained using Stochastic Gradient Descent with Momentum (SGDM) as the optimiser with a batch size of 8, $1 \times$ e-8 as learning rate, for ten epochs. All the algorithms are executed using Python code in the Google-Colab platform in T4-GPU as the runtime. Table 1 lists all the hyperparameters tuned for the proposed study.

3.7. Statistical analysis

The current study utilizes a student t-test to assess the statistical significance between the SIFT features extracted from the early-stage ROP and the normal images. The Paired t-test with 95% CI is used to estimate the significance of variance calculated between the features extracted. The statistical analyses were executed using IBM-SPSS application (version 29).

4. Results and discussion

The proposed work segments the ridge or demarcation lines from early-stage ROP retinal images using Swin

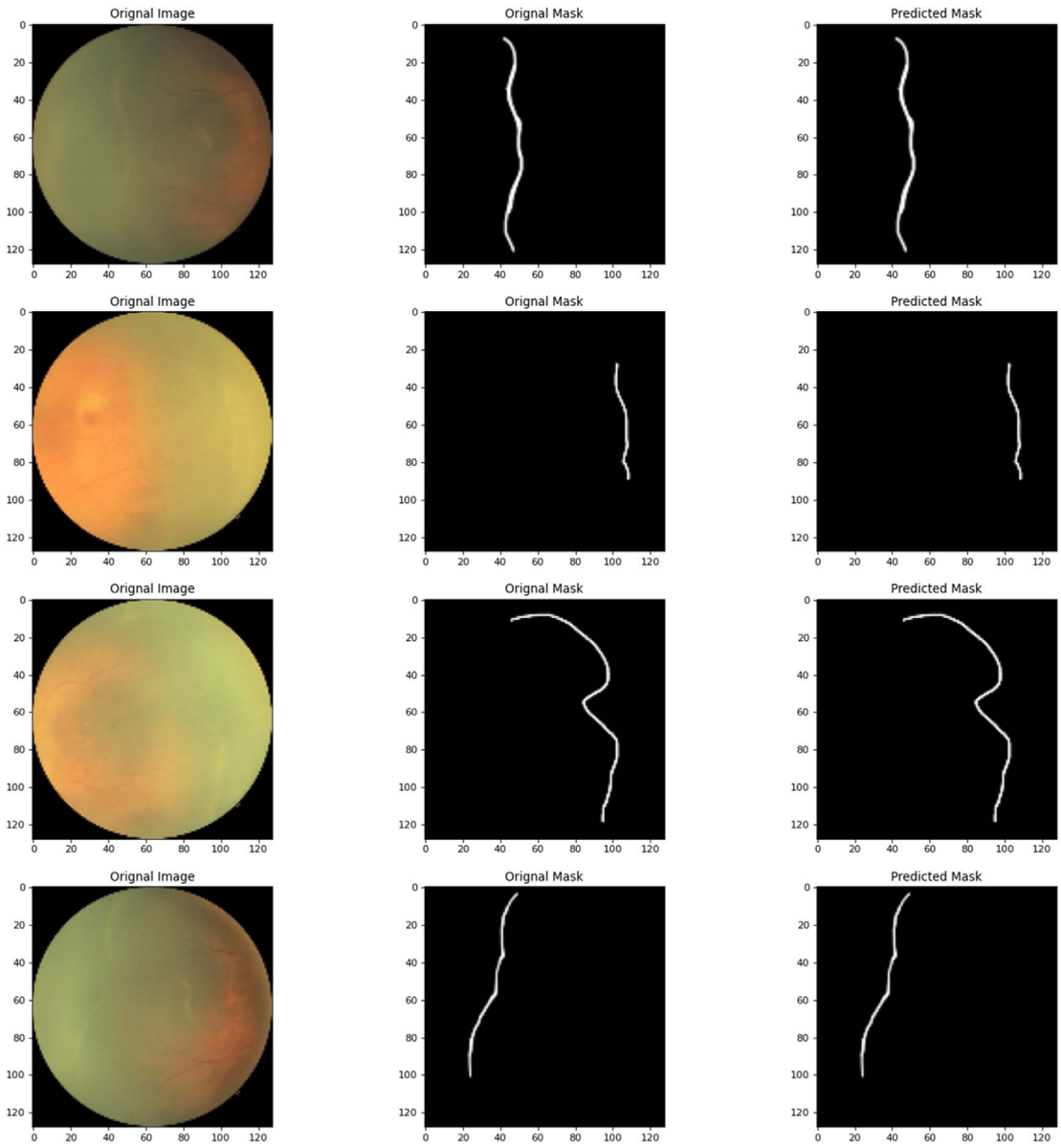


Figure 4. Results of segmentation using swin U-Net CNN.

U-Net CNN, which is trained and tested in a publicly available dataset, HVDPDROP. The qualitative results of the segmentation when compared to the labelled images are depicted in figure 4. The predicted image has the segmented demarcation line or ridge from the raw fundus images. These results are found to replicate the annotations provided. The performance of the Swin U-Net in segmentation of ridges and demarcation lines in early-stage ROP is given in table 2. Figure 5 displays the results of segmented demarcation lines and ridges from the 200 images collected in real time.

The performance metrics such as accuracy, precision, recall, dice coefficient and Intersection over Union (IoU) are utilised to measure the efficiency of the Swin U-Net CNN in segmenting demarcation lines or ridges. Accuracy indicates the overall performance of segmentation, calculating the proportion of the correctly classified pixels to the overall pixels. Precision estimates the correctness of the ridges segmented, calculated by the proportion of correctly predicted ridge pixels to the number of pixels predicted as ridges. The recall ratio is the number of correctly recognised ridge pixels to the number of pixels annotated as ridges.

The demarcation line or ridge segmentation accuracy using Swin U-Net is 89.7% with 80.5% precision, 92.6% recall, 75.76% IoU, and 0.86 as the Dice coefficient. This demonstrates that most Positive regions, such as demarcation lines and ridges, are segmented correctly.

Figures 6 and 7 show the confusion matrices obtained from the classification of early-stage ROP using ML classifiers such as SVM, RF and *k*-NN, with extracted 2000 SIFT features, PCA-based 500 principal components, GLCM and ridge extracted 7 features

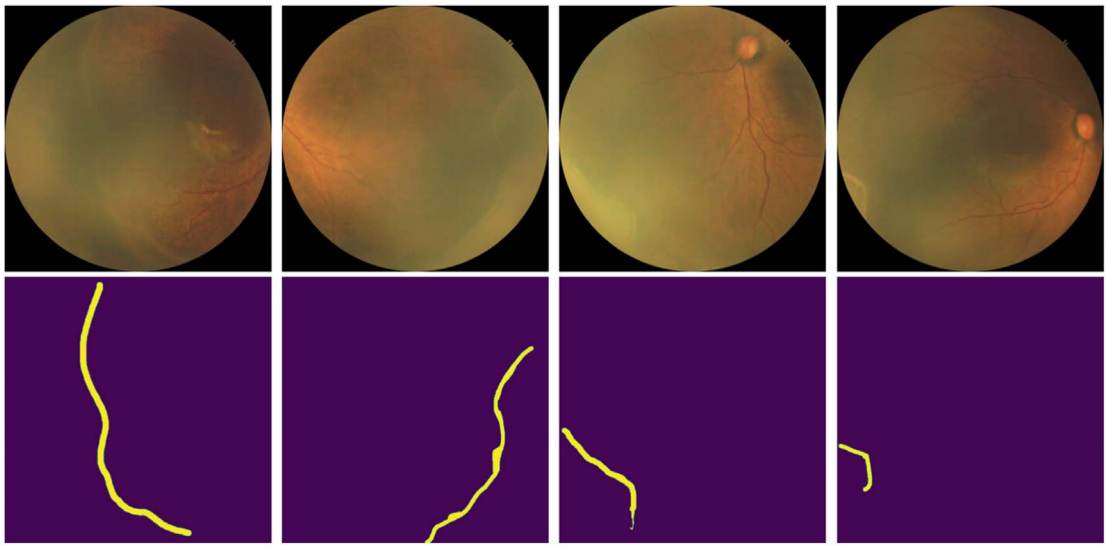


Figure 5. Segmented demarcation lines and ridges from the retinal images collected in real-time.

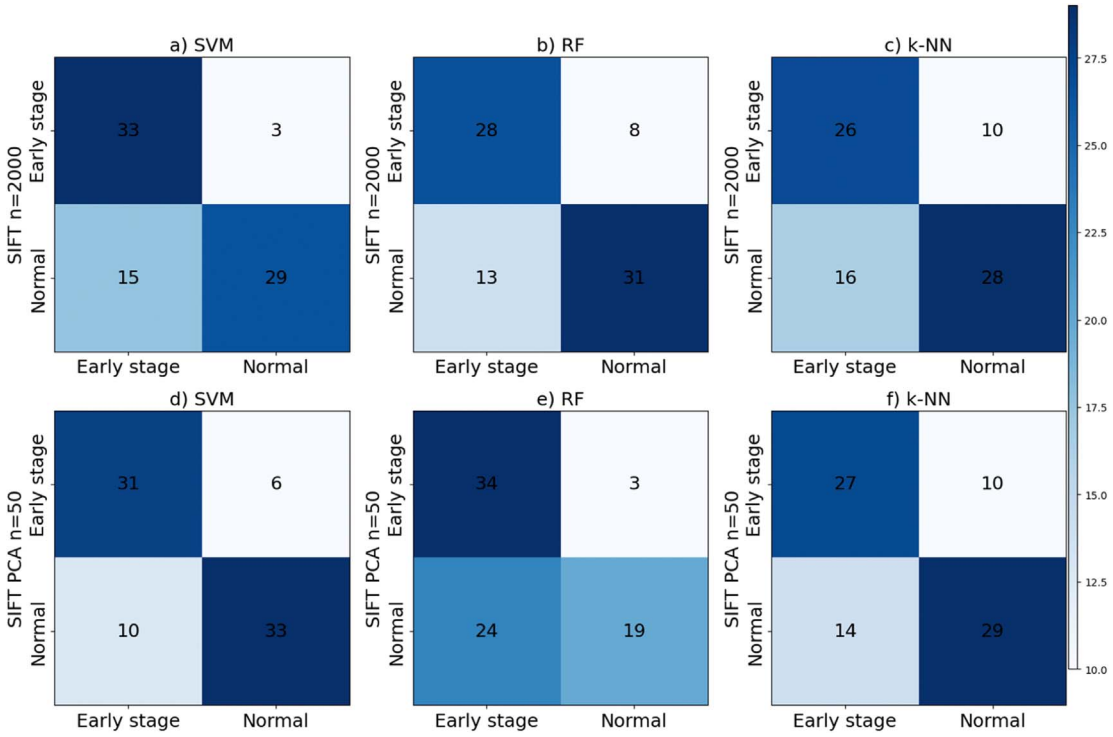


Figure 6. Confusion matrices obtained from the classification of ML classifiers (a) SVM using 2000 SIFT features; (b) RF using 2000 SIFT features; (c) *k*-NN using 2000 SIFT features; (d) SVM using 50 principal components; (e) RF using 50 principal components; (f) *k*-NN using 50 principal components.

and fused 57 using PCA-SIFT, GLCM and ridge length features. Based on the obtained confusion matrix, the performance metrics of classification, such as accuracy, sensitivity, and specificity, are estimated.

The efficiency of the ML classifiers in predicting early-stage ROP with and without PCA is compared in table 3. It was observed that the SVM classifier with PCA reduced 50 features predicts early-stage ROP with an accuracy of 80%, with 75.61% sensitivity, 84.62% specificity, 83.78% precision, 76.74% NPV and 0.8 as the Area under the curve (AUC). Whereas before dimensionality reduction, the performance of the SVM classifier using 2000 features was estimated to be 77.5% accuracy with a sensitivity of 68.75%, specificity of 90.63%, precision of 91.67%, NPV of 65.91% and an AUC of 0.79. The RF classifier predicts early-stage ROP with an accuracy of 73.5% with 68.29% sensitivity, 79.49% specificity, 77.78% precision,

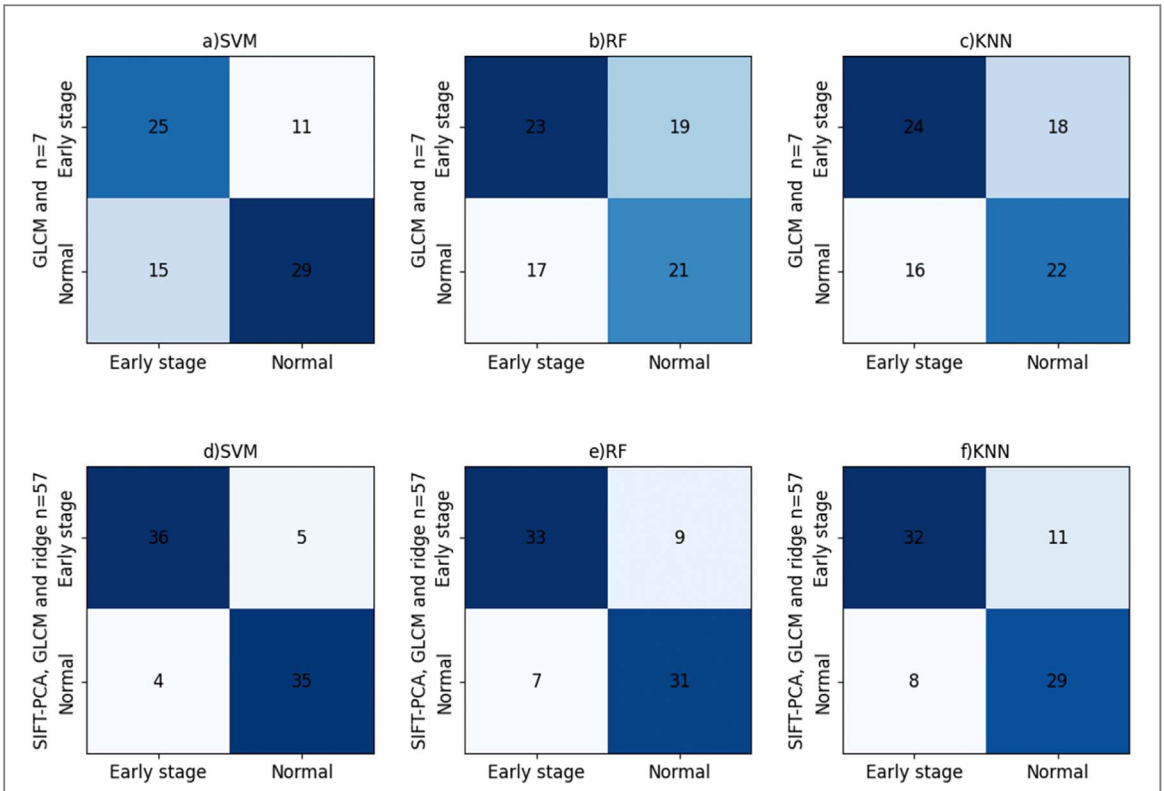


Figure 7. Confusion matrices obtained from the classification of ML classifiers using PCA-SIFT, GLCM and ridge length features (a) SVM using 7 GLCM and ridge length features; (b) RF using 7 GLCM and ridge length features; (c) k-NN using 7 GLCM and ridge length features; (d) SVM using 57 fused features; (e) RF using 57 fused features; (f) k-NN 57 fused features.

Table 2. Quantitative analysis of ridge or demarcation segmentation using swin U-net.

Metrics	Validation	Testing
True Positive	4735	5005
True Negative	10205	8935
False Positive	1083	1206
False Negative	361	394
Accuracy (%)	91.2	89.7
Precision (%)	81.4	80.5
Recall (%)	92.9	92.6
Dice coefficient	0.87	0.86
IoU (%)	76.7	75.76

70.45% NPV, and 0.74 AUC with 2000 SIFT features. Following dimensionality reduction, the accuracy of the RF classifier reduces to 66.25% with sensitivity, specificity, precision, NPV, and AUC as 58.62%, 86.36%, 91.89%, 44.19%, and 0.68, respectively. *k*-NN classifier classifies the early-stage ROP with an accuracy of 67.5% with 2000 SIFT features, which is improved to 70% using PCA with 50 features. It is seen that PCA enhances the accuracy by about 3.23% and 3.70% in SVM and *k*-NN classifiers, respectively.

Seven GLCM and ridge length features predict early-stage ROP with a maximum accuracy of 67.5%, sensitivity of 62.5%, specificity of 72.5%, precision of 69.44%, NPV of 65.91% and an AUC of 0.69 using SVM classifier. Whereas the fused 57 SIFT-PCA, GLCM, and ridge features classify early-stage ROP with an accuracy of 88.75%, sensitivity of 90%, specificity of 87.5%, precision of 87.81%, NPV of 89.74% and an AUC of 0.91. Although when the GLCM and ridge length features are utilised alone, it predicts early-stage ROP with an accuracy of 67.5%, which is less than the accuracy of 80% from the high-level 50 SIFT-based features. However, when the GLCM and ridge length features are fused with the high-level 50 SIFT-based features, the classification accuracy is improved to 88.75%. This indicates that the ROP-specific features, along with the SIFT feature descriptors, could improve the classification accuracy of early-stage ROP.

P-value estimated from the student t-test is found to be significant between the features extracted from early-stage ROP and normal images using SIFT with 2000 features ($2.35 \times e\text{-}5 < 0.05$) and SIFT-PCA with 50 features ($2.11 \times e\text{-}15 < 0.05$). The t-test results indicate that the ridge/demarcation extracted SIFT features are highly significant when dimensionally reduced using PCA.

Figure 8 depicts the AUC-ROC score plotted for the ML classifiers for early-stage ROP prediction. ROC curves were generated for SVM, Random Forest and *k*-NN classifiers with and without PCA by the false positive and true positive rates from the classification results. Based on the ROC curve, the SVM classifier achieves an AUC of 0.80 and 0.79, 0.69 and 0.91 for SIFT features with and without dimensionality reduction, GLCM-ridge and PCA-SIFT with GLCM-ridge features, respectively. The RF classifier achieves an AUC of 0.74 using 2000 SIFT features and 0.68 using

Table 3. Results of the ML classifiers in the classification of early-stage ROP from normal retinal images.

Input	Classifier	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	NPV (%)	AUC
SIFT Features = 2000	SVM	77.5	68.75	90.63	91.67	65.91	0.79
	Random Forest	73.5	68.29	79.49	77.78	70.45	0.74
	k-NN	67.5	61.9	73.68	72.22	63.64	0.68
SIFT-PCA Features = 50	SVM	80	75.61	84.62	83.78	76.74	0.8
	Random Forest	66.25	58.62	86.36	91.89	44.19	0.68
	k-NN	70	65.85	74.36	72.97	67.44	0.7
GLCM and ridge Features = 7	SVM	67.5	62.5	72.5	69.44	65.91	0.69
	Random Forest	55	57.5	52.5	54.76	55.26	0.58
	k-NN	57.5	60	55	57.14	57.89	0.60
SIFT-PCA, GLCM and ridge Features = 57	SVM	**88.75**	**90**	**87.5**	**87.81**	**89.74**	**0.91**
	Random Forest	80	82.50	77.5	78.57	81.57	0.82
	k-NN	76.25	80	72.5	74.41	78.37	0.76

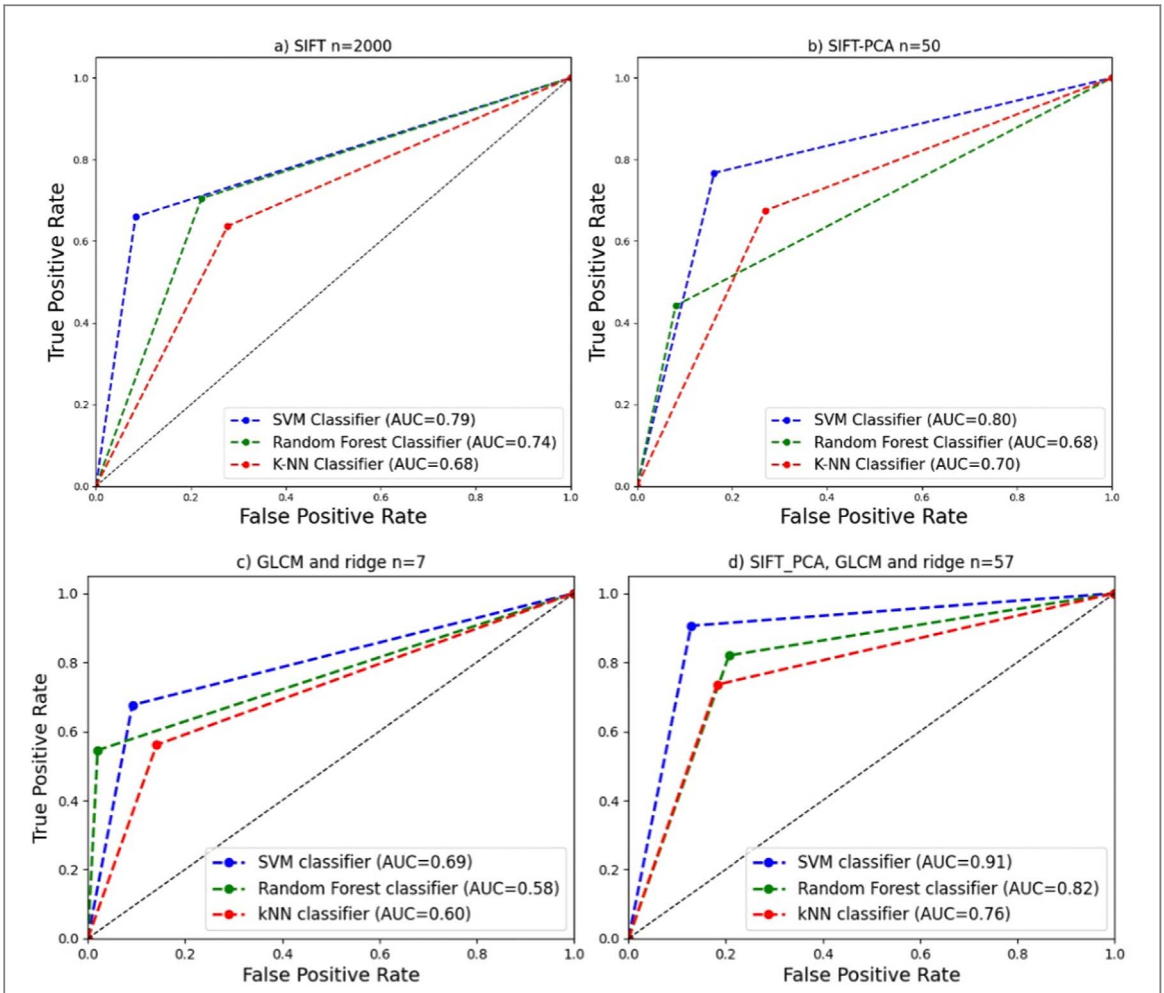


Figure 8. ROC curve plotted from the early-stage ROP classification using ML classifiers.

Table 4. Performance of the pre-trained CNN in the classification of early-stage ROP from normal retinal images.

CNN	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	NPV (%)	AUC
ResNet50	89.5	87.5	91.5	91.1	88	0.92
ShuffleNet V2	84.5	96	73	78	94.8	0.88
EfficientNet	84	79	89	87.8	80.9	0.87
MobileNet	80	65.5	94.5	92.3	73.3	0.83
VGG16	78.2	73.5	83	81.2	75.8	0.8
DarkNet19	52	40.5	63.5	52.6	51.6	0.54

50 principal components, 0.58 for GLCM-ridge and 0.82 for PCA-SIFT with GLCM-ridge features. k-NN classifier improves the AUC from 0.68 to 0.70 using PCA. The results indicate that using SIFT-PCA along with the ROP-specific features, the SVM classifiers can precisely distinguish between the early-stage ROP and normal neonatal fundus images with exceptional efficacy.

Figure 9 displays the confusion matrices obtained from the classification of early-stage ROP using six pre-trained networks, namely, ResNet50, ShuffleNet V2, EfficientNet, MobileNet, VGG16, and DarkNet19. Based on the results of confusion matrices, the performance metrics are estimated as tabulated in table 4. ResNet50 has the best classification performance with 89.5% accuracy, 87.5% sensitivity, 91.5% specificity, 91.1% precision, 88% NPV, and 0.92 AUC. The other CNN models such as ShuffleNet V2, EfficientNet b0, MobileNet V2, VGG16, and DarkNet19, achieve an accuracy of 84.5%, 84%, 80%, 78.2%, and 52%, respectively.

Figure 10 illustrates the ROC plot generated from the performance of pre-trained CNNs in classifying early-stage ROP. Based on the ROC curve, the pre-trained networks, namely ResNet50, ShuffleNet V2, EfficientNet, MobileNet, VGG16, and DarkNet19, achieve AUC of 0.92, 0.88, 0.87, 0.83, 0.8, and 0.54, respectively. The estimated AUC values reveal that the ResNet50 network is more capable of distinguishing early-stage ROP from normal preterm retinal images.

The proposed system segmented the demarcation line and ridges of early-stage ROP with an accuracy of

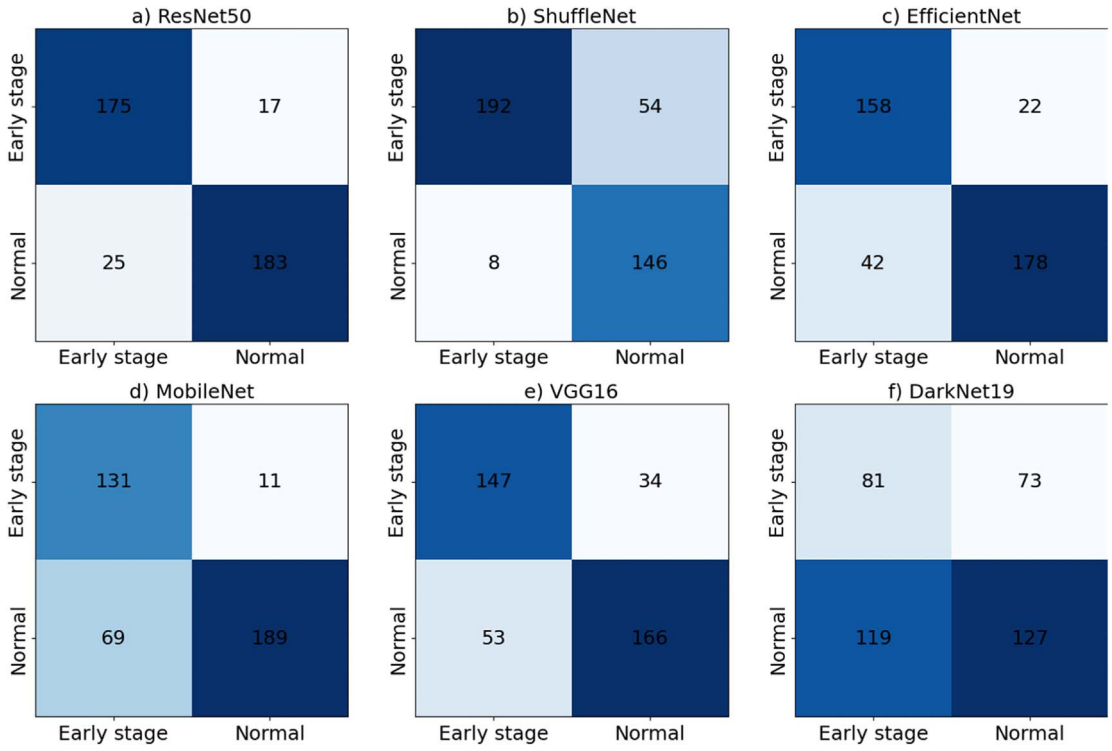


Figure 9. Confusion matrices obtained from the classification using pre-trained CNNs (a) ResNet50; (b) ShuffleNet; (c) EfficientNet; (d) MobileNet; (e) VGG16; (f) DarkNet19.

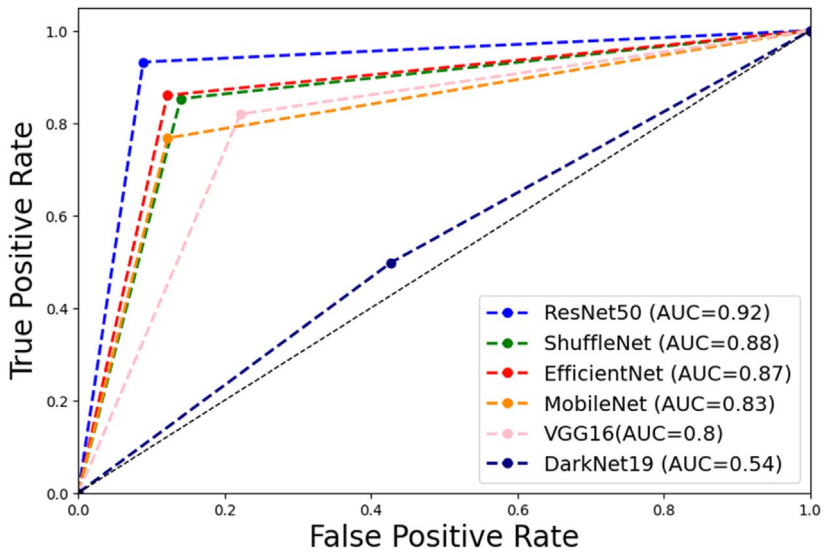


Figure 10. ROC curve plotted from the early-stage ROP classification using pre-training classifiers.

89.7%, which is comparable to state-of-the-art methodologies used by Mulay *et al* [19]. Also, they suggest that using artificial intelligence along with pre-processing the images by normalisation could improve the accuracy of detecting early-stage ROP. Kumar *et al* [33] segmented blood vessels using U-Net CNN with a dice coefficient of 0.64 to diagnose ROP with an accuracy of 88.23%. However, the Swin U-Net in the current study segments ridges and demarcation lines with a higher dice coefficient of 0.86, the accuracy obtained from ROP diagnosis is 88.75%. Hence, the classification of blood vessels along with the ridge and demarcation lines using Swin U-Net may improve the performance of early-stage ROP classification.

Feng *et al* [34] classified the stages of ROP using a semi-supervised approach using two consistency regularisation strategies. The model achieved classification accuracy of 87.31% with AUC of 0.87. Although the proposed system achieves higher classification performance, the semi-supervised model developed by Feng

et al utilises only 50% of labelled data, decreasing manual annotation costs. The pre-trained classifier Resnet50 outperformed the other methodologies in the current study in detecting early-stage ROP, which is comparable with the Faster RCNN network utilised by Tong *et al* [18].

5. Conclusion

The current study predicts early-stage ROP from the preterm fundus images using ML classifiers and deep learning networks. The characteristic demarcation lines and ridges of the early-stage ROP are segmented utilising a novel Swin U-Net. 2000 SIFT features are extracted from the segmented images and are dimensionally reduced to 50 features using PCA. Six GLCM and ridge-length features are extracted from the segmented image and are fused with the 50 PCA reduced SIFT descriptors to obtain a 57-length feature vector. Finally, the ML networks, such as SVM, Random Forest, and k-NN, are used to classify the 57 features to predict the early-stage ROP from Normal images. On the other hand, the raw retinal images are classified directly into normal and early-stage ROP using six pre-trained classifiers, namely ResNet50, ShuffleNet V2, EfficientNet, MobileNet, VGG16, and DarkNet19. The proposed work has certain limitations. Firstly, the early-stage ROP is predicted in the current study using ML classifiers based on the ridges and demarcation lines segmented from the neonatal fundus images. However, the network predicts the other stages of ROP as normal since it has no characteristic ridges or lines. Hence, the network must be trained with an improved dataset with images of all stages of ROP to have a multiclass classification. The number of images used to train and test the segmentation networks is less, reducing the performance of SwinUNet. Hence, future work can involve developing a more extensive database of neo-fundus images and their corresponding annotations. Also incorporating the features from the segmented blood vessels will improve the classification performance. Moreover, our proposed ROP diagnosing method could be employed as an economical and accessible neonatal eye care device in remote regions with inadequate healthcare services.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.1016/j.dib.2023.109839.

Author statement

The first author made significant contributions in design, experimentation, execution, data gathering, analysis, and manuscript drafting.

The second author made significant contributions in design, drafting, and editing of the manuscript.

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Ethical approval

The research was conducted in accordance with the principles embodied in the Declaration of Helsinki and in accordance with local statutory requirements. All participants gave written informed consent to participate in the study. The proposed work is permitted by IEC, SRM Hospital & Medical College and Research Centre, with number 8241/IEC/2022.

Conflicts of interest

All the authors have no conflicts of interest.

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